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Perspective

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Trends in testing effect research: from lab to classroom, but not yet for all learners

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Active retrieval leads to better learning outcomes than passive study. This perspective, in which we analyze 23,850 publications, reveals an increasing focus on the educational applications of this testing effect but limited attention to subpopulations with learning disabilities. Using dyslexia as a case study, we identify theoretical grounds to question whether testing effects generalize universally, highlighting the need for empirical research. Future research should examine benefits of testing in neurodiverse learners to develop tailored interventions.

Testing effect research: from laboratory to the classroom

Actively attempting to retrieve to-be-learned information from memory, rather than passively reviewing it, has been shown to consistently enhance subsequent memory retrieval performance—a phenomenon known as the *testing effect* [e.g., see refs. 1–4]. The first systematic experimental investigations are often traced to[5] and[6], who demonstrated that active recall improved retention compared to passive rereading. Since then, the effectiveness of testing or retrieval practice as a learning method has been demonstrated in both controlled laboratory studies and classroom environments across a (neuro)diverse range of subjects [for reviews, see refs. 7–9]. Research has also demonstrated that the benefits of testing extend beyond simple retention to include the transfer and application of knowledge to new contexts and problem types[10,11], making it particularly valuable for real-world learning outcomes. This extensive body of research highlights the potential of retrieval practice as a powerful learning tool, both in formal educational contexts and through digital platforms like Babbel (https://babbel.com), Duolingo (https://duolingo.com), MemoryLab (https://memorylab.nl/en) and Quizlet (https://quizlet.com) which are collectively used by hundreds of millions of learners around the world [also see ref. 7].

Because retrieval practice is widely applied in educational and digital learning contexts, it is important to consider whether research findings generalize across diverse learner populations. In other words, to what extent do existing studies account for the heterogeneity of real-world learners? Notably, the benefits of testing effects hinge on learners' ability to successfully recall the target information — a requirement that may be challenging for low-performing students or those with learning disabilities. This work aims to examine how individual differences and neurodiversity, specifically in relation to specific learning disabilities, have been addressed in retrieval practice research and whether adaptations may be needed to demonstrate or increase its effectiveness for all learners. Note that this perspective does not aim to provide a comprehensive theoretical or mechanistic account of the testing effect (for which we refer the reader to works of[12,13]. Rather, arguments grounded in theoretical background and methodological approaches are presented together to characterize the latest trends in the literature, and to highlight the current limitations and gaps. More detailed descriptions of the methods employed (such as thematic keyword- and co-citation analyses) are available in the Supplementary information.

Trends

Understanding trends in testing effect research requires an overview of its coverage in academic output over time. We conducted a literature search for journal publications covering testing effects, retrieval practice, or test-enhanced learning, identifying 1774 relevant studies (see Supplementary Fig. 1 for a schematic depiction of the methodological steps taken in this and the following analyses, and see Supplementary Note 1 and Supplementary Table 1 for detailed information about the literature search). Figure 1a shows that research on the testing effect has steadily increased since the early 1990s, peaking at around 150 publications in 2022. To assess whether publications primarily focus on theoretical mechanisms or educational applications of the testing effect, we examined the 20 journals with the most publications, categorizing them as "basic research," "applied research," or "mixed" based on their stated editorial focus. Figure 1b illustrates the absolute number of papers published in each category per year. The data indicate that the number of mixed and applied research publications has grown substantially over the past two decades (see Supplementary Note 2 and Supplementary Table 2 for more details). Conversely, the number of publications in journals focused on fundamental research has declined since 2012. Figure 1C illustrates the declining *proportion* of basic research over time.

Thematic analysis

Next to the analysis based on editorial statements, we also categorized the 1774 papers using keyword-based network analysis, a data-driven approach. To that end, we identified all noun phrases from the abstracts and titles of a set of publications, and filtered out general terms (such as "conclusion", "study"). The network analysis then clusters the noun phrases based on co-

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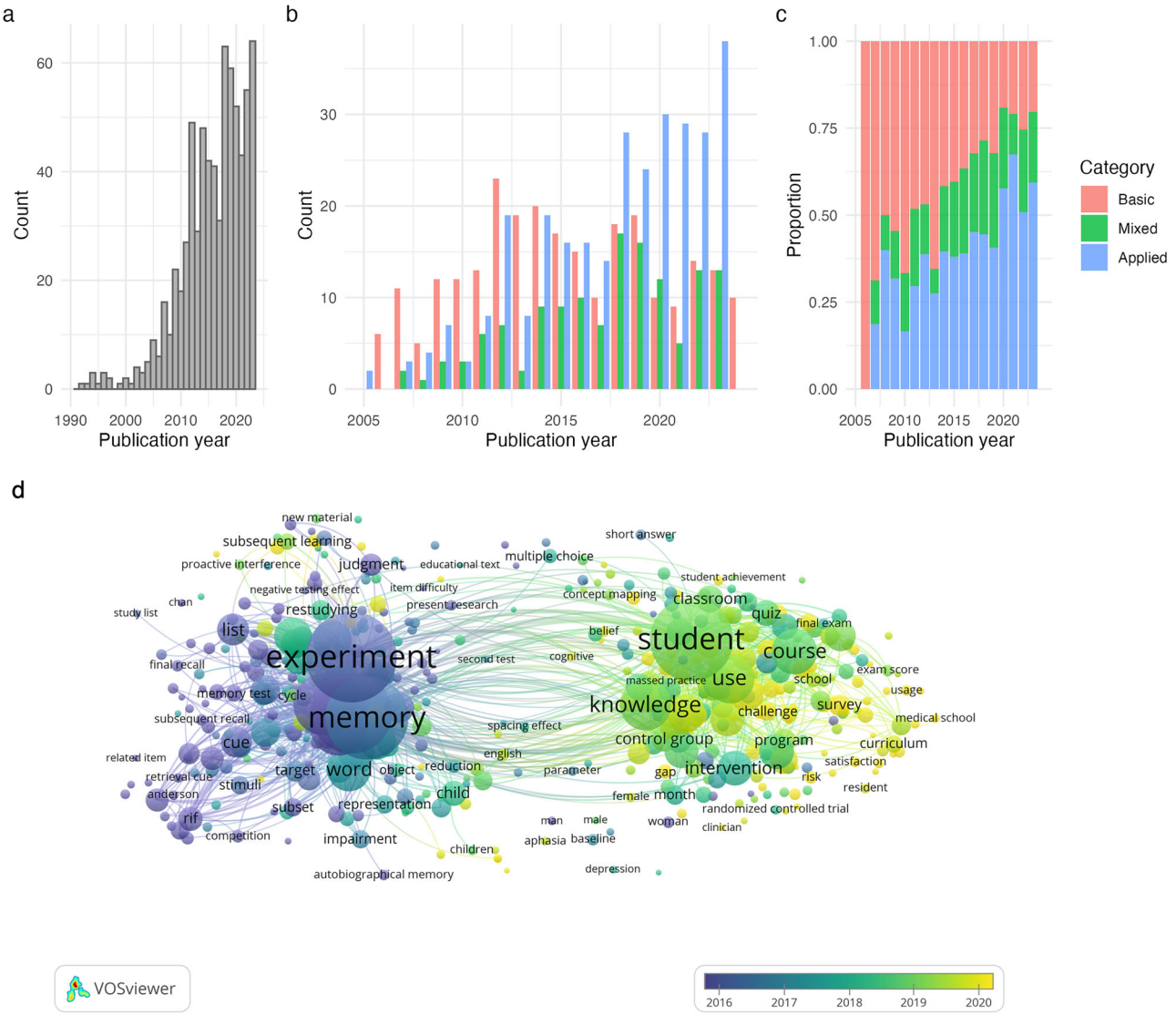


Fig. 1 | Annual trends and thematic analysis of research on the testing effect. **a** Presents the yearly count of academic journal publications on the testing effect from 1990 to 2024. **b** Displays the annual number of papers published in journals that, according to their editorial statements, emphasize basic or fundamental research over the past 20 years. Panel **c** illustrates the *proportion* of papers published in journals categorized as focusing on basic, mixed, or applied research on the testing effect. Finally, **d** visualizes a keyword-based network analysis of all studies on the testing effect, with colors indicating the year of publication. Dots represent extracted noun phrases, with dot size corresponding to the number of research items in which the phrase appears. The distance between any two dots reflects the frequency of co-occurrence of the corresponding noun phrases within a document. More details on the methods are described in Supplementary Notes 1–3.

occurrence (see Supplementary Note 3 for a more detailed description of the noun-phrase selection and clustering procedure). If, over a larger set of papers, terms are frequently used in the same paper, they are considered representative of the core concepts or focus points of a research field[14–16]. Figure 1d shows the results of a thematic analysis of 453 noun phrases extracted from the abstracts and titles of the papers. Two main clusters of noun phrases could be identified, corresponding to two semantic topics covered by the field. Cluster 1 (left) contains 217 noun phrases mainly relating to experimental or fundamental studies in retrieval practice literature (e.g., "memory", "experiment", "cue", "stimuli"). Cluster 2 (right), forming a separate body of terms, contained 200 noun phrases, mainly related to more applied questions in retrieval practice literature (e.g., "student", "classroom", "knowledge" and "exam score",). The terms in Fig. 1d are colored by the average publication date associated with the occurrence of the noun phrase. A clear shift from fundamental to applied research over time is visible: Research on applications of the testing effect (Cluster 2) has, on average, been published more recently than the fundamental studies on the testing effect (Cluster 1). To summarize, both analyses reveal a clear shift towards greater emphasis on (educational) applications and a decline in focus on fundamental research.

Individual differences and learning disabilities

Whereas participants in lab studies are typically drawn from homogeneous populations (e.g., students enrolled in a university program), educational practice serves students with far more diverse backgrounds and learning needs. Therefore, it is critical to carefully consider whether individual differences between learners are taken into account when applying conclusions from lab-based testing effect research to educational practice. Moreover[17], and[18] show that the vast majority of research on the testing effect considers the effect at the group level, potentially obscuring the question whether retrieval practice might be detrimental for some learners. In general, some studies reviewed by[17] support the idea that the *magnitude* of the testing effect varies substantially between individuals, but the studies differ in the degree to which they identify consistent moderators of this variability. Some studies report systematic associations with cognitive factors such as general intelligence, attentional control, or working memory capacity[8,19,20], while others

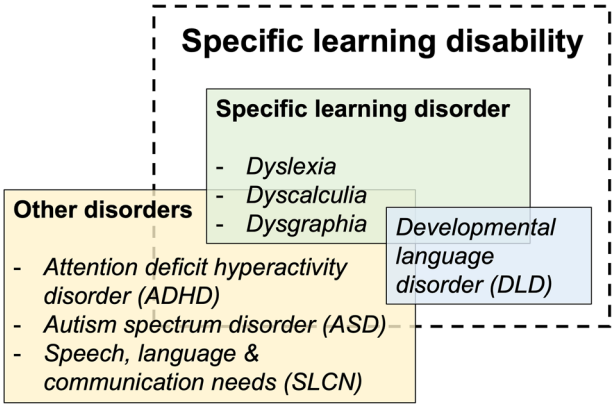


Fig. 2 | A taxonomy of specific learning disabilities. Specific learning disabilities (SLDs) encompass a group of specific learning disorders that impede the acquisition of foundational academic skills, including dyslexia (difficulties with word recognition and spelling), dyscalculia (difficulties with math), and dysgraphia (difficulties with writing). Developmental Language Disorder (DLD) also often leads to significant academic challenges and is treated here within the broader category of SLDs. Other neurodevelopmental conditions, such as ADHD, Autism Spectrum Disorder (ASD), and Specific Language Impairment (SLI), may also affect educational outcomes but are not classified as primary SLDs as they may also affect areas outside academic skills.

do not report or find reliable predictors of individual differences in testing effect magnitude[21–24].

Notably[18], conducted a comprehensive scoping review examining 20 individual difference constructs across 20 studies and found that the majority reported no significant relationship between individual differences and testing effect magnitude. However, they argue that these null findings should be interpreted cautiously due to methodological limitations including insufficient statistical power, low reliability of measures, and restriction of range in samples. Critically[18], conclude that it would be premature to interpret these findings as evidence that individual differences do not influence testing effects. Instead, they propose that while the testing effect likely operates for most learners, optimal implementation conditions may vary across individuals, and that future research should examine interactions between individual differences and contextual factors. This perspective aligns with our argument: the fundamental mechanisms may be universal, but their operation may be compromised or require adaptation in learners with specific learning disabilities.

One obvious source of individual variability in learning processes and outcomes is *Specific Learning Disability* (SLD, see Fig. 2). SLD refers to a variety of conditions that cause persistent difficulties in acquiring and using academic skills [see Fig. 2,[25,26]]. Specific learning *disorders* include conditions that underlie these ongoing challenges in acquiring basic academic skills, such as efficient phonological processing, word recognition, and spelling (dyslexia), writing (dysgraphia), and mathematical calculations (dyscalculia). Developmental Language Disorder (DLD) is a distinct underlying condition that can also cause SLD, and mainly impairs the *implicit* learning of language (as opposed to learning from explicit academic instruction that is affected in specific learning disorders). Other disorders, such as attention disorders or autism spectrum disorders, are sometimes also mentioned as a cause of SLDs. However, since they do not primarily impact learning itself, we do not include these disorders in this overview. In the following, we will use the term SLDs to refer to educational difficulties caused by specific learning disorders or developmental language disorder.

Learners with SLD generally have normal intelligence, functional hearing, and functional vision[27,28]. Their performance can be characterized by inaccurate or slow reading, skipping lines when reading aloud, frequent spelling errors, illegible handwriting, and difficulties with basic arithmetic tasks. The academic hurdles caused by these difficulties prevent them from attaining academic levels that align with their intellectual abilities[29]. SLD is estimated to affect approximately 5–15% of school-aged children[26,27,29]. Developmental dyslexia is the most common subtype of SLD, affecting approximately 80% of children identified with SLD[29]. Given the high rates of occurrence, it is crucial that the effects of SLD on the testing effect is assessed before it is applied in the classroom.

Co-citation analysis

To examine the extent to which testing effects have been studied for individuals with SLDs specifically, we conducted an additional literature search that included research on testing effects *and* SLDs (see Supplementary Note 1), and we used a bibliometric analysis to explore the overlap between the two fields of research [[30], see Supplementary Note 4]. Here, we used co-citation coupling to determine the semantic relatedness of multiple sets of publications. Co-citation occurs when two articles are cited simultaneously by a subsequent publication[31]. Frequent co-citation implies that the documents share related themes or topics, and a lack of co-citation implies that two research collections are independent of each other[30].

Figure 3 shows the co-citation network for a total of 23,850 journal articles that matched the testing effect search criteria, the specific learning disabilities criteria, or both. It provides a visual map of how studies on retrieval practice and specific learning disabilities are interlinked within the literature. The size of each node shows its relative influence (based on total citations), and proximity between nodes reflects how often those studies are cited together, highlighting thematic connections. The analysis reveals 19 main clusters of frequently co-cited articles. A list with the numeric results of this analysis, as well as thematic interpretations for these clusters, is shown in Supplementary Table 3. Notably, the distinct purple cluster to the right captures a body of work dedicated to the testing effect, suggesting this area is both highly cited and relatively well-delineated from other themes in the field.

In total, we identified 204 research items that were either (a) part of the testing effect research cluster and co-cited with papers in any of the SLD clusters, or (b) part of any of the SLD clusters and co-cited with papers in the retrieval practice cluster. However, just eight publications out of these 204 research items were (a) original research articles, reporting (b) an experimental manipulation to compare an active retrieval to a passive study/review condition in (c) a specific subgroup of SLD learners. Interestingly, all of these studies covered learners with DLD; we found *no* work covering learners with dyslexia, dyscalculia, or dysgraphia. It is worth noting that while comorbidity between DLD and other SLDs can occur, none of the eight DLD studies identified in our analysis reported comorbid SLDs in their participant samples. Moreover, even if undetected comorbidity were present in some cases, this limited body of work remains insufficient to draw conclusions about how testing effects operate across the diverse range of cognitive profiles represented by different SLDs. The list of 191 remaining studies consisted of research in two main categories: (1) educational interventions for students with SLDs or investigations into how SLDs affect memory performance without explicit testing effect manipulations; and (2) studies examining testing effects through mechanisms commonly implicated in SLDs (e.g., attention, working memory capacity), but exclusively using typically developing samples. Critically, the co-citation analysis highlighted a notable gap: only very few studies directly investigate the mechanisms or efficacy of testing compared to passive study in SLD populations.

As reported above, we identified just eight studies with (a) a testing effect manipulation in (b) an SLD group: learners with DLD. Note that while our focus has been on the standard testing effect paradigm (i.e., comparing an active retrieval condition to a passive study condition) which allows for a clear theoretical interpretation of underlying mechanisms, we emphasize that applied research comparing real-world educational interventions with and without retrieval practice components is equally important. When using this more lenient exclusion criterion, no additional studies were identified. A short summary of these studies can be found Supplementary Note 5 and Supplementary Table 4. Overall, they provide valuable insights in the mechanisms underling testing benefits in DLD, and suggest that

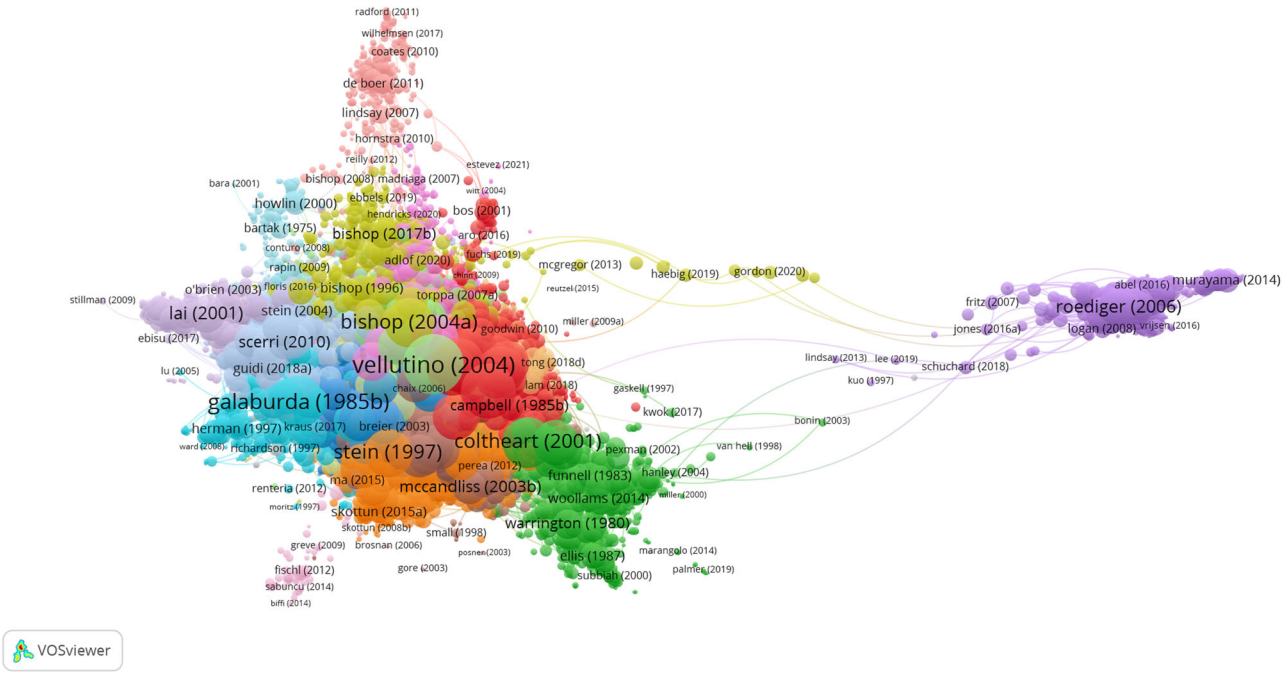


Fig. 3 | Bibliometric co-citation coupling analysis of studies on the retrieval practice benefit and on specific learning disabilities. Each node in the network represents an individual publication, with node size proportional to its total number of citations. The spatial proximity of nodes reflects their degree of co-citation, with closer nodes representing studies more frequently cited together. The node colors denote distinct thematic clusters within the literature. The well-delineated purple cluster to the right comprises studies focusing on the testing effect, indicating this as a distinct and highly cited area of inquiry within the broader research landscape. More details on the co-citation method employed can be found in Supplementary Note 4.

retrieval practice can support word learning and retention in these learners. However, the evidence base remains limited: few studies are available, existing studies have very small sample sizes (7–16 participants) and most are conducted outside the classroom (see Supplementary Table 4). In addition, the authors of this work explicitly point to the fact that mechanisms that drive the testing effect might differ in learners with DLD, due to the unique challenges they face in learning[32]. For example[33], demonstrated that when learning verbs, children with DLD often struggle to link verbs to their correct complement[34,35], potentially due to their inherent relational and less imageable character than nouns[36]. Accordingly[33], found that while retrieval practice benefited typically developing children, leading to better recall immediately and after a week, children with DLD showed a smaller advantage, with no benefit after a week. On a positive note[37], did find benefits of retrieval practice in learners with DLD on other materials, such as nouns.

The authors point to several mechanisms that may cause retrieval practice to operate differently in this population compared to typical learners. They note that the desirable difficulty principle—whereby retrieval should be effortful enough to promote learning but not so difficult as to cause failure—may have a different threshold for children with DLD due to their encoding deficits and working memory limitations. Specifically, the optimal level of retrieval difficulty may be lower for DLD learners, meaning that spacing intervals requiring greater contextual reinstatement might exceed their cognitive capacity more quickly than for typically developing peers, potentially negating the benefits of retrieval practice. Overall, although valuable first steps for learners with DLD are taken, the co-citation analysis reveals strikingly little overlap between testing effects and SLD research.

The lack of research on other SLDs (e.g., dyslexia, dyscalculia) might be taken as evidence that these conditions do not meaningfully modulate testing effects. However, we will discuss the mechanisms implicated in developmental dyslexia in more detail below, and argue that this lack of evidence cannot be interpreted as evidence of universal applicability.

Developmental dyslexia: a case study

The term *developmental dyslexia* is commonly used to denote a developmental problem in which a learner's reading performance is substantially below the level expected given the learner's general intelligence level and age. Dyslexia is characterized by—given typical educational opportunities and intact intellectual and sensory abilities—unexpected difficulties in reading[38]. The disorder, which has a strong genetic component[39], has been estimated to affect about 5–10% of children[40,41].

Learners with dyslexia have difficulties with written-word recognition and phonological decoding (using letter-sound mapping knowledge to decode novel words) which may result from underlying phonological impairments [e.g., see refs. 42–44]. Dyslexia has been associated with a number of other cognitive deficits, including working memory[45,46], executive functions[47–49], motor function[50], semantic integration[51], implicit sequence learning[52,53], and artificial grammar learning[54]. Consequently, a dyslexia diagnosis suggests that a variety of cognitive deficits could interfere with reading and phonological decoding.

To understand how the testing effect might differ for learners with dyslexia compared to typically developing learners, it is important to consider the mechanisms that have been proposed to underlie this phenomenon. One key theoretical account is the *elaborative retrieval hypothesis*[55], which draws from associative memory theories[56,57]. According to this hypothesis, active retrieval enhances learning by promoting *semantic elaboration*: when retrieving information, learners activate and integrate cue-relevant knowledge, leading to enriched memory traces[3]. A separate but complementary framework is the *desirable difficulty* account[58], which underlies the *retrieval effort hypothesis*[59,60]. This framework posits that retrieval activities that are effortful—but still successful—enhance long-term retention more than easy or passive learning tasks. Thus, while elaborative retrieval emphasizes the formation of rich semantic connections, the desirable difficulty framework focuses on the role of retrieval effort in optimizing learning. In the following sections, we will explore how both semantic elaboration and retrieval effort may operate differently in learners with dyslexia, complicating generalization of testing effect findings from

typically developing samples. Other theoretical accounts have also been proposed to explain the testing effect, including transfer-appropriate processing[61,62], and episodic context theories[12]. While these frameworks provide valuable insights into retrieval-based learning, we focus here on elaborative retrieval and desirable difficulty because these mechanisms most directly implicate cognitive processes that are well-documented to be impaired in dyslexia.

First, the elaborative retrieval hypothesis centers on the concept of *spreading activation*, whereby active retrieval not only reinforces existing retrieval cues but also promotes the formation of new semantic associations by activating related knowledge structures. This process enhances recall for both the tested material and semantically related content[63].

Importantly, research indicates that associative memory processes involving semantic processing may be impaired or delayed in learners with dyslexia. For example, in reading paradigms, numerous neuroimaging (EEG/MEG) studies report that readers with dyslexia show reduced or delayed event-related potentials—such as the N400 response, a marker of semantic incongruence—when reading semantically incongruent sentences[51,64–67]. Similar findings have been obtained in auditory paradigms, suggesting that the semantic processing deficit may extend beyond written processing[68,69]. Other researchers have argued that these reduced brain potentials arise from difficulties with sensory- or written text processing, rather than semantic integration per se[70]. Regardless of whether the bottleneck occurs at the level of low-level sensory processing or higher-level semantic integration, it is likely that the associative spreading activation processes that support elaborative retrieval are compromised in at least some learning contexts. If so, this could attenuate one of the key mechanisms that underpins the testing effect in typical learners.

Second, according to the desirable difficulty framework, the benefits of retrieval practice depend critically on the level of cognitive effort involved: retrieval should be sufficiently effortful to promote learning but not so difficult that it leads to failure[59,71]. This leads to an *inverse-U shaped* relationship between task difficulty and learning gains[72]. Optimal difficulty levels appear to differ between learners with and without dyslexia. For instance[73], found that increasing difficulty in a word-learning task improved learning outcomes for skilled spellers but impaired learning for less skilled spellers—a pattern consistent with lower optimal difficulty thresholds in dyslexic learners. Similarly[74], found that adapting learning material to match both the dyslexia type and reading skill level of Arabic learners with dyslexia resulted in significantly better learning outcomes compared to providing non-adapted standard material—a pattern consistent with the importance of matching material difficulty to learners' current abilities and specific learning needs. Parallel findings have been observed in other domains[75]: reported that reducing item exposure enhanced learning in typical readers but not in learners with dyslexia, and[76] found that higher-order rule learning tasks impaired performance in dyslexic learners while benefiting typical ones.

Beyond the process of retrieval itself, the standard *format* of retrieval practice tasks may pose additional barriers for learners with dyslexia. Retrieval exercises typically require reading written prompts and producing typed responses—activities that demand extra effort due to decoding and orthographic processing difficulties[38,43]. This increased effort in accessing and responding to retrieval cues, combined with frustration over spelling errors and slower response times, may reduce engagement and motivation. In adaptive learning systems that adjust difficulty based on speed and accuracy, learners with dyslexia may be penalized for their decoding or typing difficulties rather than their actual retrieval capabilities, potentially leading to task avoidance. Thus, even if the core retrieval mechanisms remain functional, practical barriers in task format may prevent learners with dyslexia from engaging effectively with retrieval practice [e.g., see ref. 77].

Together, these results suggest that learners with dyslexia may exceed the optimal difficulty threshold more readily when retrieval practice is made more demanding, which could negate its intended benefits. Whether these cognitive differences actually reduce testing effects, eliminate them, or simply require adaptations in implementation, remains an open empirical question.

Implications beyond dyslexia

Overall, the above-mentioned evidence suggests that the mechanisms driving the testing effect—semantic elaboration and effortful retrieval—may function differently in learners with developmental dyslexia. Although outlining the specific learning mechanisms impacted in all SLDs is beyond the scope of this perspective, we would like to note that similar differences may apply to other specific learning disabilities. For example, increased cognitive demands inherent to dyscalculia or dysgraphia—originating from persistent deficits in working memory, symbolic mapping, or orthographic coding—may undermine the principle of effortful retrieval and desirable difficulties (in dyscalculia, visuospatial working memory limitations interfere with numerical representation and problem solving, reducing the capacity to handle additional retrieval demands[78,79]. Dysgraphia involves effortful motor sequencing and orthographic processing that strain transcription and encoding, potentially overwhelming learners during retrieval practice[80]). At a minimum, these ideas underscore the urgent need for explicit research to examine whether the cognitive mechanisms underlying the testing effect operate similarly across SLDs—and, if not, to understand how they differ and what implications this has for selecting materials, formats, and implementation strategies. As our bibliometric analysis shows, such research is currently lacking, highlighting the need for inclusive studies to investigate the benefits and limitations of retrieval-based learning across diverse SLD profiles. Without such empirical work, the assumption that retrieval practice universally enhances learning risks overlooking critical neurocognitive differences.

Perspective

A better understanding of the mechanisms that underlie retrieval-based learning in heterogeneous populations, including learners with SLDs, could lead to more tailored educational interventions. Such interventions (e.g., computerized learning applications) can be adapted to the needs of students with SLDs in different ways. For example, presenting specific sets of materials, such as balancing the proportion of verbs and nouns in a study set to manage task difficulty [see ref. 35] should lead to more efficient testing-based learning for learners with DLD. Making learning environments more inclusive can also entail attending to how the format of responses affects learners. For example, recent work by ref. 77 used an adaptive retrieval practice paradigm with both typing-based and speech-based response modes to compare learning outcomes between dyslexic and typically developing learners. The results showed that learners with dyslexia recalled more words when using speech than typing, suggesting that removing a bottleneck caused by phonological decoding and orthographic mapping can make retrieval practice more effective for this group. Mathematical modeling of the response times revealed that the typing-specific disadvantage for dyslexic learners was primarily due to processing delays, rather than in memory per se, highlighting a key opportunity to make the software more inclusive by optimizing its input and interaction design. Similar adjustments—such as providing spoken prompts, accepting verbal responses, or integrating auditory feedback—can help reduce demands from decoding and orthographic processing and better support neurodiverse learners. Other examples might include the customization of pacing/pausing options, scaffolding, or adapting content and repetition schedules to the needs of individual learners. A systematic analysis of how each of these features affects learners' motivational, affective, cognitive, and perceptual needs can guide the design of inclusive retrieval practice environments that are both accessible and effective for a wide range of learners, including those with SLDs [e.g., see ref. 81].

It is worth noting that this paper is intended as a perspective piece rather than a systematic review. As such, our literature search was conducted exclusively within the Web of Science database, which was selected for its breadth and quality, reference export options, as well as its significant overlap with other databases such as PsycINFO and PubMed[82]. In addition,

our approach identifies research gaps based on established connections between fields through patterns of which papers are cited together, rather than through exhaustive examination of individual publications. This means that individual papers that are not well integrated into the main citation networks of either the testing effect literature or the SLD literature would be unlikely to appear in our analysis. Such papers might be isolated due to publication in specialized journals with limited readership, use of non-standard terminology, recency (very recent work that has not yet accumulated any citations).

Although this approach does likely not capture every relevant publication, it allows for the identification of key trends and gaps in the literature —such as the shift from fundamental to applied research and the paucity of studies focusing on learners with specific learning disabilities. We acknowledge that future work employing more systematic searches across multiple databases would be a valuable next step. Such efforts would also enable a more comprehensive examination of the underlying mechanisms in other neurodiverse profiles beyond specific learning disabilities. In this paper, we examine specific learning disorders (dyslexia, dyscalculia, dysgraphia) and developmental language disorder; other conditions such as ADHD and ASD, while relevant to learning, were beyond the scope of the current paper due to its focus on education-specific disabilities, but warrant further investigation in future studies.

In conclusion, a growing body of literature on the testing effect underscores its importance in enhancing long-term retention. In recent years, the main focus of research on the testing effect has shifted from the mechanisms underlying the testing effect to applying it in educational practice. Although the cognitive mechanisms that drive the testing effect may be significantly impacted in learners with SLDs, strikingly little work has studied the benefits of testing in learners with SLDs. This gap in the literature is particularly important given the widespread prevalence of SLDs, including dyslexia. Accordingly, this work leads to two key recommendations. First, there is a need for greater attention to individual differences in testing effect research. While the testing effect has shown promise when averaged over (homogeneous) samples of participants, very few studies report whether it consistently benefits all learners, or whether it may also negatively impact some individuals. Second, there is a need for research into the mechanisms that underlie retrieval-based learning in specific subgroups of learners with SLDs. Ultimately, a better understanding of these mechanisms will lead to the development of inclusive and effective applications that employ the benefits of testing in educational practice.

Data availability

Data for this project can be found on the Open Science Framework: https://osf.io/cz6th/.

Code availability

Analyses scripts for this project can be found on the Open Science Framework: https://osf.io/cz6th/.

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References

1. Glover, J. A. The" testing" phenomenon: not gone but nearly forgotten. *J. Educ. Psychol.* **81**, 392 (1989).
2. Karpicke, J. D. & Aue, W. R. The testing effect is alive and well with complex materials. *Educ. Psychol. Rev.* **27**, 317–326 (2015).
3. Carpenter, S. K., Pan, S. C. & Butler, A. C. The science of effective learning with spacing and retrieval practice. *Nat. Rev. Psychol.* **1**, 496–511 (2022).
4. Carpenter, S. K. Encouraging students to use retrieval practice: a review of emerging research from five types of interventions. *Educ. Psychol. Rev.* **35**, 96 (2023).
5. Abbott, E. E. On the analysis of the factor of recall in the learning process. *Psychol. Rev. Monogr. Suppl.* **11**, 159 (1909).
6. Gates, A. Recitation as a factor in memorizing. *Arch. Psychol.* **40**, 104 (1917).
7. Agarwal, P. K., Nunes, L. D. & Blunt, J. R. Retrieval practice consistently benefits student learning: a systematic review of applied research in schools and classrooms. *Educ. Psychol. Rev.* **33**, 1409–1453 (2021).
8. Moreira, B. F. T., Pinto, T. S. S., Starling, D. S. V. & Jaeger, A. Retrieval practice in classroom settings: a review of applied research. in *Frontiers in Education*, Vol. 4, 5 (Frontiers Media SA, 2019).
9. Roediger, H. L. & Butler, A. C. The critical role of retrieval practice in long-term retention. *Trends Cogn. Sci.* **15**, 20–27 (2011).
10. Carpenter, S. K. Testing enhances the transfer of learning. *Curr. Dir. Psychol. Sci.* **21**, 279–283 (2012).
11. Pan, S. C. & Rickard, T. C. Transfer of test-enhanced learning: meta-analytic review and synthesis. *Psychol. Bull.* **144**, 710 (2018).
12. Karpicke, J. D., Lehman, M. & Aue, W. R. Retrieval-based learning: an episodic context account. in *Psychology of learning and motivation*, Vol. 61, 237–284 (Elsevier, 2014).
13. McDermott, K. B. Practicing retrieval facilitates learning. *Annu. Rev. Psychol.* **72**, 609–633 (2021).
14. Cheng, Q., Wang, J., Lu, W., Huang, Y. & Bu, Y. Keyword-citation-keyword network: a new perspective of discipline knowledge structure analysis. *Scientometrics* **124**, 1923–1943 (2020).
15. Cross, R., Borgatti, S. P. & Parker, A. Making invisible work visible: Using social network analysis to support strategic collaboration. *Calif. Manag. Rev.* **44**, 25–46 (2002).
16. Wasserman, S. & Faust, K. Social network analysis: Methods and applications (Cambridge University Press, 1994).
17. de Lima, M. F. R. & Buratto, L. G. Retrieval practice effect and individual differences: current status and future directions. *J. Cogn. Psychol.* **36**, 443–456 (2024).
18. Smith-Peirce, R. N. & Butler, A. C. A scoping review of research on individual differences in the testing effect paradigm. *Learn. Individ. Differ.* **118**, 102602 (2025).
19. Brewer, G. A. & Unsworth, N. Individual differences in the effects of retrieval from long-term memory. *J. Mem. Lang.* **66**, 407–415 (2012).
20. Bertilsson, F., Stenlund, T., Wiklund-Hörnqvist, C. & Jonsson, B. Retrieval practice: Beneficial for all students or moderated by individual differences? *Psychol. Learn. Teach.* **20**, 21–39 (2021).
21. Minear, M., Coane, J. H., Boland, S. C., Cooney, L. H. & Albat, M. The benefits of retrieval practice depend on item difficulty and intelligence. *J. Exp. Psychol. Learn. Mem. Cogn.* **44**, 1474 (2018).
22. Agarwal, P. K., Finley, J. R., Rose, N. S. & Roediger III, H. L. Benefits from retrieval practice are greater for students with lower working memory capacity. *Memory* **25**, 764–771 (2017).
23. Yang, C. et al. Do working memory capacity and test anxiety modulate the beneficial effects of testing on new learning? *J. Exp. Psychol.: Appl.* **26**, 724 (2020).
24. Zheng, Y., Sun, P. & Liu, X. L. Retrieval practice is costly and is beneficial only when working memory capacity is abundant. *npj Sci. Learn.* **8**, 8 (2023).
25. Fletcher, J. M. Classification and identification of learning disabilities. *Learn. Learn. Disabil.* **4**, 1–26 (2012).
26. Grigorenko, E. L. et al. Understanding, educating, and supporting children with specific learning disabilities: 50 years of science and practice. *Am. Psychol.* **75**, 37 (2020).
27. Lagae, L. Learning disabilities: definitions, epidemiology, diagnosis, and intervention strategies. *Pediatr. Clin. North Am.* **55**, 1259–1268 (2008).
28. Shaywitz, S. E. Dyslexia. *N. Engl. J. Med.* **338**, 307–312 (1998).
29. Karande, S. & Kulkarni, M. Specific learning disability: the invisible handicap. *Indian Pediatr.* **42**, 315–319 (2005).
30. McBurney, M. K. & Novak, P. L. What is bibliometrics and why should you care? in *Proc. IEEE International Professional Communication Conference*, 108–114 (IEEE, 2002).

31. Small, H. Co-citation in the scientific literature: a new measure of the relationship between two documents. *J. Am. Soc. Inf. Sci.* **24**, 265–269 (1973).
32. Leonard, L. B. & Deevy, P. Retrieval practice and word learning in children with specific language impairment and their typically developing peers. *J. Speech Lang. Hear. Res.* **63**, 3252–3262 (2020).
33. Leonard, L. B. et al. Learning verbs in sentences: Children with developmental language disorder and the role of retrieval practice. *J. Speech Lang. Hear. Res.* **67**, 4446–4465 (2024).
34. Owen Van Horne, A. J. & Lin, S. Cognitive state verbs and complement clauses in children with sli and their typically developing peers. *Clin. Linguist. Phon.* **25**, 881–898 (2011).
35. Leonard, L. B. et al. Production operations contribute to the grammatical morpheme limitations of children with specific language impairment. *J. Mem. Lang.* **43**, 362–378 (2000).
36. Ma, W., Golinkoff, R. M., Hirsh-Pasek, K., McDonough, C. & Tardif, T. Imageability predicts the age of acquisition of verbs in Chinese children. *J. Child Lang.* **36**, 405–423 (2009).
37. Leonard, L. B., Christ, S. L., Deevy, P., Karpicke, J. & Kueser, J. B. Retrieval practice and word learning by children with developmental language disorder: Does expanding retrieval provide additional benefit? *J. Speech Lang. Hear. Res.* **67**, 1530–1547 (2024).
38. Lyon, G. R., Shaywitz, S. E. & Shaywitz, B. A. A definition of dyslexia. *Ann. dyslexia.* **53**, 1–14 (2003).
39. Scerri, T. S. & Schulte-Körne, G. Genetics of developmental dyslexia. *Eur. Child Adolesc. Psychiatry* **19**, 179–197 (2010).
40. Wagner, R. K. et al. The prevalence of dyslexia: a new approach to its estimation. *J. Learn. Disabil.* **53**, 354–365 (2020).
41. Yang, L. et al. Prevalence of developmental dyslexia in primary school children: a systematic review and meta-analysis. *Brain Sci.* **12**, 240 (2022).
42. Navas, A. L. G. P., Ferraz, É. d. C. & Borges, J. P. A. Phonological processing deficits as a universal model for dyslexia: evidence from different orthographies. in *CoDAS*, Vol. 26, 509–519 (SciELO Brasil, 2014).
43. Pennington, B. F., Van Orden, G. C., Smith, S. D., Green, P. A. & Haith, M. M. Phonological processing skills and deficits in adult dyslexics. *Child Dev.* **61**, 1753–1778 (1990).
44. Tijms, J. Verbal memory and phonological processing in dyslexia. *J. Res. Read.* **27**, 300–310 (2004).
45. Jeffries, S. & Everatt, J. Working memory: Its role in dyslexia and other specific learning difficulties. *Dyslexia* **10**, 196–214 (2004).
46. Smith-Spark, J. H. & Fisk, J. E. Working memory functioning in developmental dyslexia. *Memory* **15**, 34–56 (2007).
47. Helland, T. & Asbjørnsen, A. Executive functions in dyslexia. *Child Neuropsychol.* **6**, 37–48 (2000).
48. Reiter, A., Tucha, O. & Lange, K. W. Executive functions in children with dyslexia. *Dyslexia* **11**, 116–131 (2005).
49. Varvara, P., Varuzza, C., Sorrentino, A. C., Vicari, S. & Menghini, D. Executive functions in developmental dyslexia. *Front. Hum. Neurosci.* **8**, 120 (2014).
50. Chaix, Y. et al. Motor impairment in dyslexia: the influence of attention disorders. *Eur. J. Paediatr. Neurol.* **11**, 368–374 (2007).
51. Schulz, E. et al. Impaired semantic processing during sentence reading in children with dyslexia: combined fmri and erp evidence. *Neuroimage* **41**, 153–168 (2008).
52. Hedenius, M. et al. Impaired implicit sequence learning in children with developmental dyslexia. *Res. Dev. Disabil.* **34**, 3924–3935 (2013).
53. Kelly, S. W., Griffiths, S. & Frith, U. Evidence for implicit sequence learning in dyslexia. *Dyslexia* **8**, 43–52 (2002).
54. Pavlidou, E. V., Williams, J. M. & Kelly, L. M. Artificial grammar learning in primary school children with and without developmental dyslexia. *Ann. dyslexia* **59**, 55–77 (2009).
55. Carpenter, S. K. Cue strength as a moderator of the testing effect: the benefits of elaborative retrieval. *J. Exp. Psychol. Learn. Mem. Cogn.* **35**, 1563 (2009).
56. Anderson, J. R. A spreading activation theory of memory. *J. Verbal Learn. Verbal Behav.* **22**, 261–295 (1983).
57. Anderson, J. R. & Matessa, M. A production system theory of serial memory. *Psychol. Rev.* **104**, 728 (1997).
58. Bjork, E. L. et al. Making things hard on yourself, but in a good way: creating desirable difficulties to enhance learning. *Psychol. Real World Essays Illus. Fundam. Contrib. Soc.* **2**, 56–64 (2011).
59. Rowland, C. A. The effect of testing versus restudy on retention: a meta-analytic review of the testing effect. *Psychol. Bull.* **140**, 1432 (2014).
60. Karpicke, J. D. & Blunt, J. R. Retrieval practice produces more learning than elaborative studying with concept mapping. *Science* **331**, 772–775 (2011).
61. Morris, C. D., Bransford, J. D. & Franks, J. J. Levels of processing versus transfer appropriate processing. *J. Verbal Learn. Verbal Behav.* **16**, 519–533 (1977).
62. Veltre, M. T., Cho, K. W. & Neely, J. H. Transfer-appropriate processing in the testing effect. *Memory* **23**, 1229–1237 (2015).
63. Chan, J. C., McDermott, K. B. & Roediger III, H. L. Retrieval-induced facilitation: initially nontested material can benefit from prior testing of related material. *J. Exp. Psychol. Gen.* **135**, 553 (2006).
64. Schulz, E. et al. Reading for meaning in dyslexic and young children: distinct neural pathways but common endpoints. *Neuropsychologia* **47**, 2544–2557 (2009).
65. Helenius, P., Salmelin, R., Service, E. & Connolly, J. F. Semantic cortical activation in dyslexic readers. *J. Cogn. Neurosci.* **11**, 535–550 (1999).
66. Paz-Alonso, P. M. et al. Neural correlates of phonological, orthographic and semantic reading processing in dyslexia. *NeuroImage Clin.* **20**, 433–447 (2018).
67. Wang, E. et al. Word recognition semantic processing defects in Chinese children with developmental dyslexia: an event-related potential study. *NeuroQuantology* **15**, 221−231 (2017).
68. Jednoróg, K., Marchewka, A., Tacikowski, P. & Grabowska, A. Implicit phonological and semantic processing in children with developmental dyslexia: Evidence from event-related potentials. *Neuropsychologia* **48**, 2447–2457 (2010).
69. Taylor, M. & Keenan, N. Erps to orthographic, phonological, and semantic tasks in dyslexic children with auditory processing impairment. *Dev. Neuropsychol.* **15**, 307–326 (1999).
70. Silva, P. B., Ueki, K., Oliveira, D. G., Boggio, P. S. & Macedo, E. C. Early stages of sensory processing, but not semantic integration, are altered in dyslexic adults. *Front. Psychol.* **7**, 170703 (2016).
71. Halamish, V. & Bjork, R. A. When does testing enhance retention? A distribution-based interpretation of retrieval as a memory modifier. *J. Exp. Psychol. Learn. Mem. Cogn.* **37**, 801 (2011).
72. Maddox, G. B., Pyc, M. A., Kauffman, Z. S., Gatewood, J. D. & Schonhoff, A. M. Examining the contributions of desirable difficulty and reminding to the spacing effect. *Mem. Cogn.* **46**, 1376–1388 (2018).
73. Eskenazi, M. A. & Nix, B. Individual differences in the desirable difficulty effect during lexical acquisition. *J. Exp. Psychol. Learn. Mem. Cogn.* **47**, 45 (2021).
74. Alghabban, W. G. & Hendley, R. Adaptive e-learning and dyslexia: an empirical evaluation and recommendations for future work. *Interact. Comput.* **36**, 65–74 (2024).
75. McGraw Fisher, J., Liederman, J., Johnsen, J., Lincoln, A. & Frye, R. A demonstration that task difficulty can confound the interpretation of lateral differences in brain activation between typical and dyslexic readers. *Laterality Asymmetries Body Brain Cogn.* **17**, 340–360 (2012).
76. Pavlidou, E. V. & Williams, J. M. Implicit learning and reading: Insights from typical children and children with developmental dyslexia using the artificial grammar learning (agl) paradigm. *Res. Dev. Disabil.* **35**, 1457–1472 (2014).

77. Wilschut, T., Sense, F. & van Rijn, H. Modality matters: Evidence for the benefits of speech-based adaptive retrieval practice in learners with dyslexia. *Topics in Cognitive Science* (2024).
78. Menon, V. Working memory in children's math learning and its disruption in dyscalculia. *Curr. Opin. Behav. Sci.* **10**, 125–132 (2016).
79. Attout, L. & Majerus, S. Working memory deficits in developmental dyscalculia: the importance of serial order. *Child Neuropsychol.* **21**, 432–450 (2015).
80. Aiswarya, G. & Joseph Ponniah, R. Dysgraphia and memory: insights into the cognitive mechanisms, neural correlates, and intervention strategies. *Integr. Psychol. Behav. Sci.* **58**, 1778–1792 (2024).
81. Arslan, B., Wilschut, T., van Rijn, H. & van der Velde, M. Evidence-Based Dynamic Personalization for Learning and Assessment Tools (ED-PLAT): Machine-and Learner-Driven Adaptation to Support All Learners. In International Conference on Artificial Intelligence in Education pp 477–491 (Cham: Springer Nature Switzerland, 2025).
82. Falagas, M. E., Pitsouni, E. I., Malietzis, G. A. & Pappas, G. Comparison of PubMed, Scopus, Web of Science, and Google Scholar: strengths and weaknesses. *FASEB J.* **22**, 338–342 (2008).

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TW, HvR, and FS conceptualized the project. TW conducted the literature search, bibliometric analyses, and prepared the initial draft. The final draft was collaboratively written by TW, FS, and HvR.

Competing interests

The authors declare no competing interests.

Additional information

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